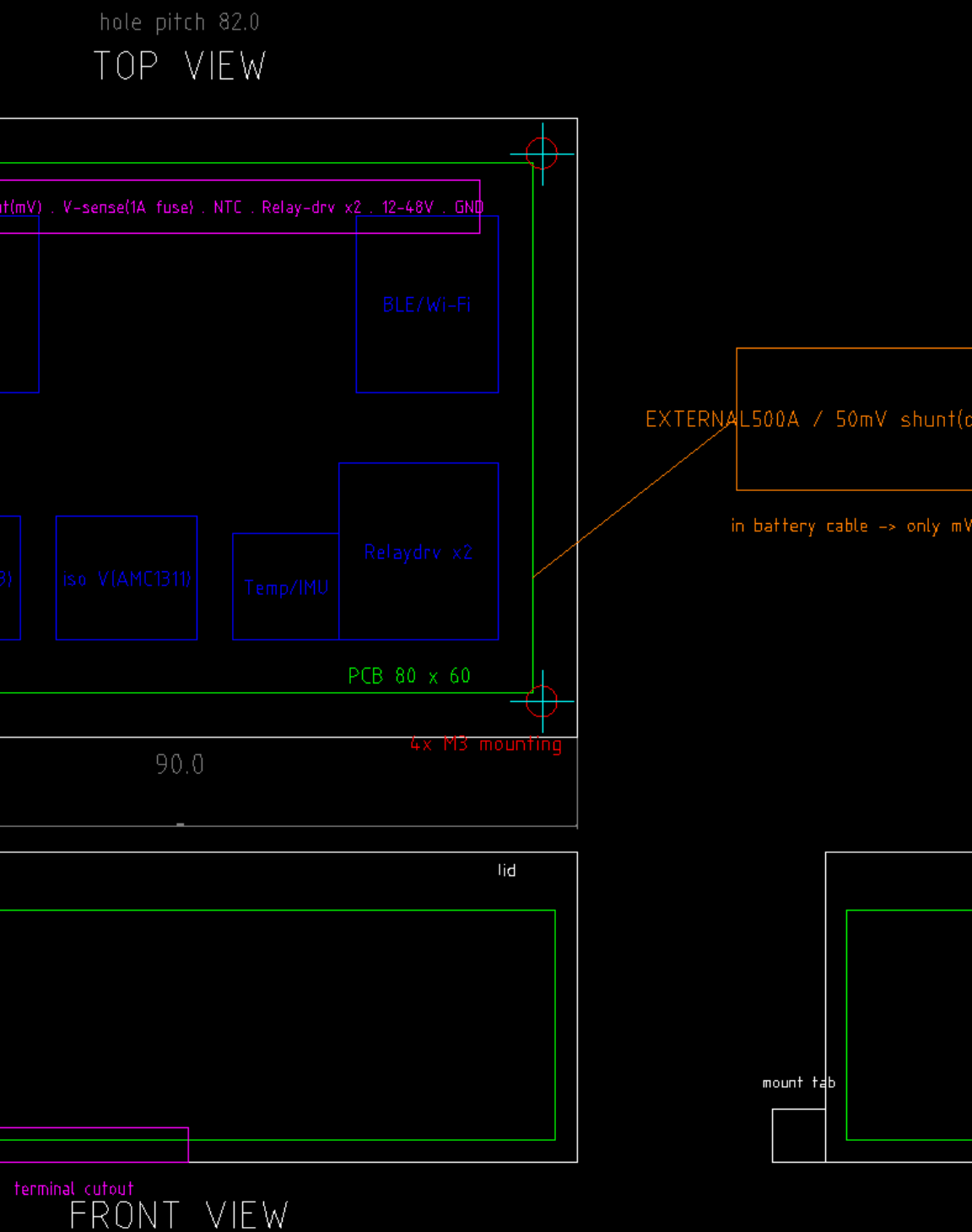




V-Guard Sentinel: the self-aware home inverter

Team: Codey Tingle (TI3271) · Ayush Manoj Garg, Eshan Shukla, Tanay Chaplot

Problem: India's home inverters switch in under 10 ms but are blind to the one part that actually fails — the battery. It dies without warning, the family loses backup and pays ₹15–18 k, and V-Guard pays inside the warranty window. Today's products sense voltage and current and turn none of it into foresight.

Module | Enclosure 90x70x35 mm ABS UL94 V-0 | PCB 80x60 | mounts ON
ing via an EXTERNAL 500A/50mV shunt or ACS758 Hall - the PCB carries only mV signal +
V-Guard Sentinel · Team Codey Tingle (TI3271) · Big Idea Tech 2026, Track 4

Our Vision & Mission

Vision

Every V-Guard product moves from smart to intelligent: it predicts its own failure, learns its owner's habits and acts on its own — offline, on a ₹300 microcontroller.

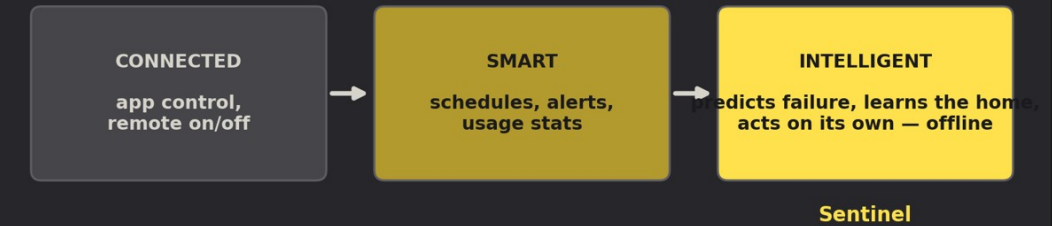
Goals

- Never let a battery die unannounced
- Keep the essentials alive through every blackout
- Turn warranty into evidence, not argument

Tech Stack

ESP32-S3 · TensorFlow Lite Micro (int8) · Kalman filtering · event-based NILM · IEEE 1159 power quality · ECDSA-signed logs · Flask dashboard · ESP-IDF firmware · Python/C prototype with 245 tests

Where V-Guard's products are today → where Sentinel takes them



~10 y → ~3.3 y

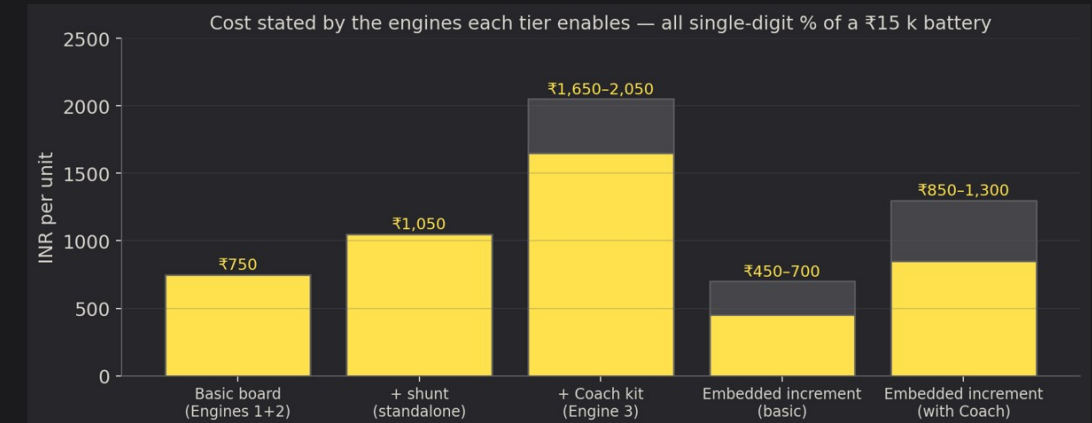
lead-acid life at 25 °C vs 41 °C:
heat halves life every +10 °C

≈1.5 %

of V-Guard revenue goes to
warranty (≈₹69 cr, FY24)

< 10 ms

V-Guard Prime transfer time —
the inverter is fast; the battery is blind



Sentinel Engine Suite

1 Engine 1 · Battery Health

Coulomb counting and a Kalman filter fuse current, voltage and temperature into state of charge; a 28 k-parameter int8 network turns each outage cycle into a state-of-health band and the warning “replace within 6–14 weeks”.

2 Engine 2 · Habit Autopilot

Learns the home's load and outage rhythm, detects a blackout in under 2 s by a 2-of-3 vote, and sheds heavy loads by tier so the fridge, lights and router last longest. Every fault fails safe: coil off means loads on.

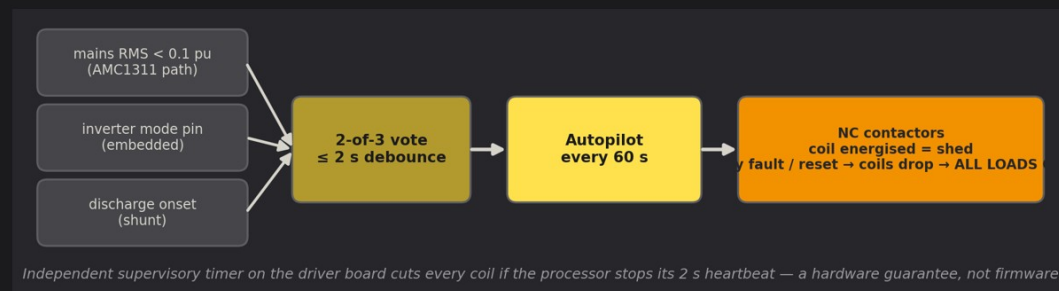
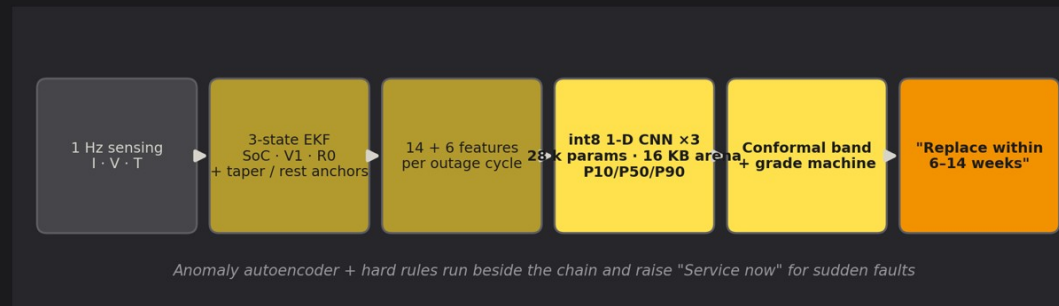
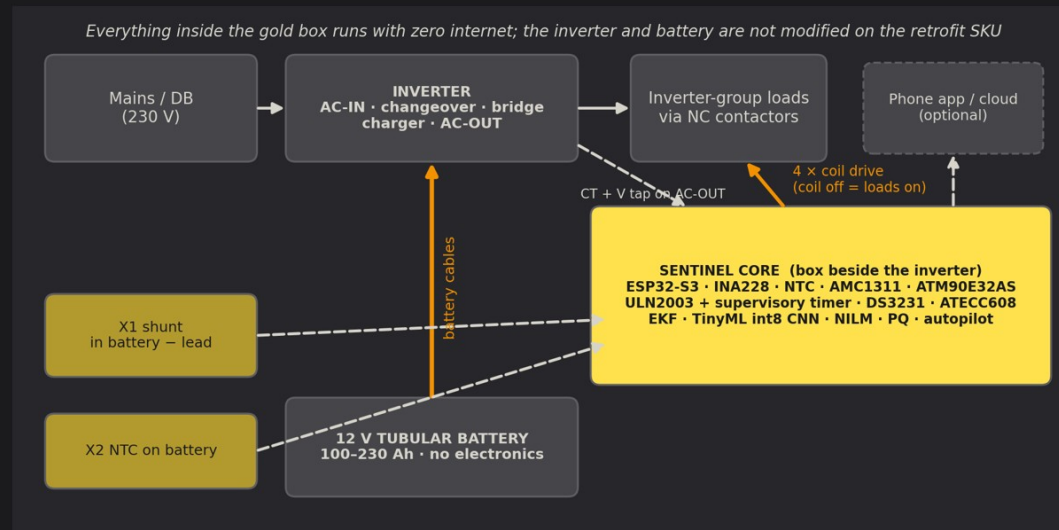
3 Engine 3 · Energy Coach & Grid Shield

A metering front-end on the inverter output gives real and reactive power. Appliance events are recognised and priced; sags, swells and interruptions are logged as appliance-protection evidence.

4 Engine 4 · Fleet Intelligence

Consented, pseudonymised per-cycle summaries (~1 KB/day) train better models centrally and return over the air. Federated learning is designed and deliberately on hold.

Key advantage: every core function runs with zero internet — a virtue in India, not a compromise; retrofits the installed base or embeds in new inverters.



The Product Kits

Sentinel-Retrofit box

Core board beside the inverter · external $0.1\text{ m}\Omega$ shunt in the battery negative lead · NTC on the battery · CT + voltage tap on AC-OUT · 4 fail-safe contactor channels.

Perfect for: the installed base — a 30–45 minute electrician install, no inverter modification

Sentinel-Embedded board

Daughter-board inside a new V-Guard inverter · Kelvin-taps the inverter's own shunt · commands the charger through a DAC on the PWM node with a hardware voltage/thermal ceiling.

Perfect for: the Smart Pro family — adaptive charging and the life-extension claim live here only

Energy Coach kit

ATM90E32AS metering front-end + split-core CT + voltage tap + isolator: real and reactive power at 3 Hz on the inverter output (optional second CT on the mains).

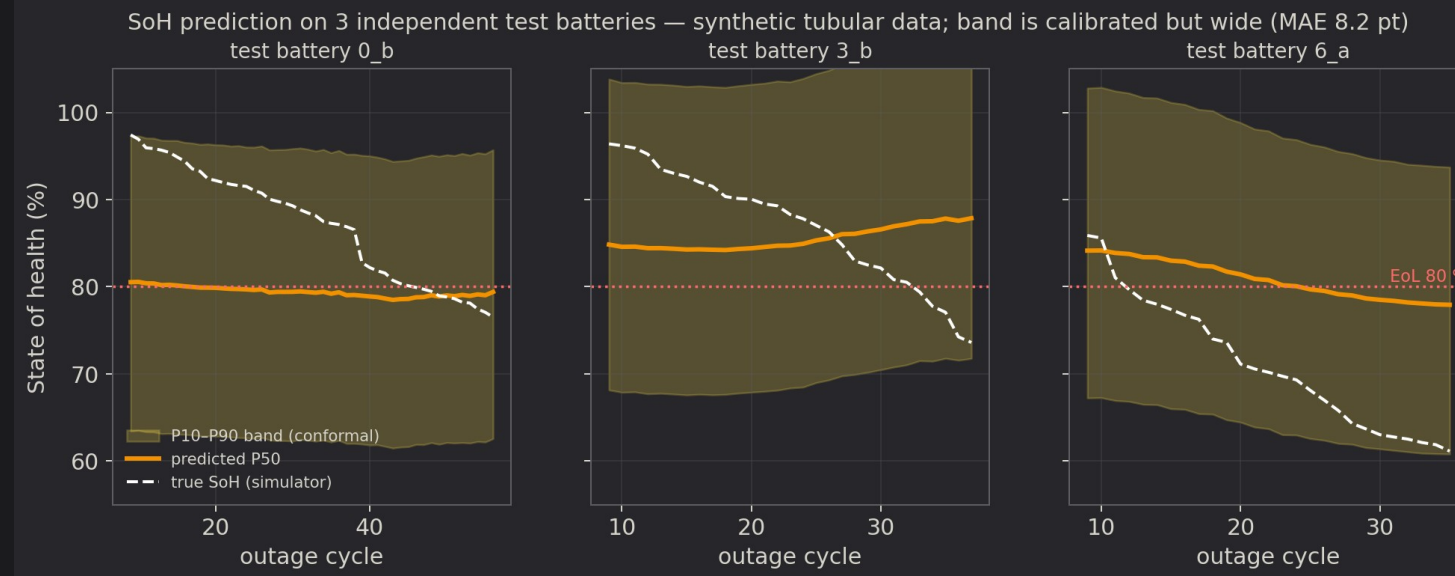
Perfect for: homes that want appliance-level kWh, ₹ at ToD tariff and grid-event evidence

Contactor panel

2–4 × 25 A normally-closed DIN contactors with 12 V coils, one hardware-locked medical channel and a supervisory timer that drops every coil if the processor stops its heartbeat.

Perfect for: tiered load-shedding — granularity equals the number of channels wired

Cost by engine: basic board $\approx$ ₹750 · with shunt $\approx$ ₹1,050 · full retrofit with Coach $\approx$ ₹1,650–2,050 · embedded increment ₹450–1,300 — single-digit % of a ₹15 k battery.



Engine 1: Predicting Battery Life

01

Estimate

A 3-state Kalman filter (SoC, polarisation, R0) on 1 Hz current, voltage and temperature. SoC re-anchors to 100 % on charge-current taper and to the OCV table only after a true rest with the charger off.

02

Reduce

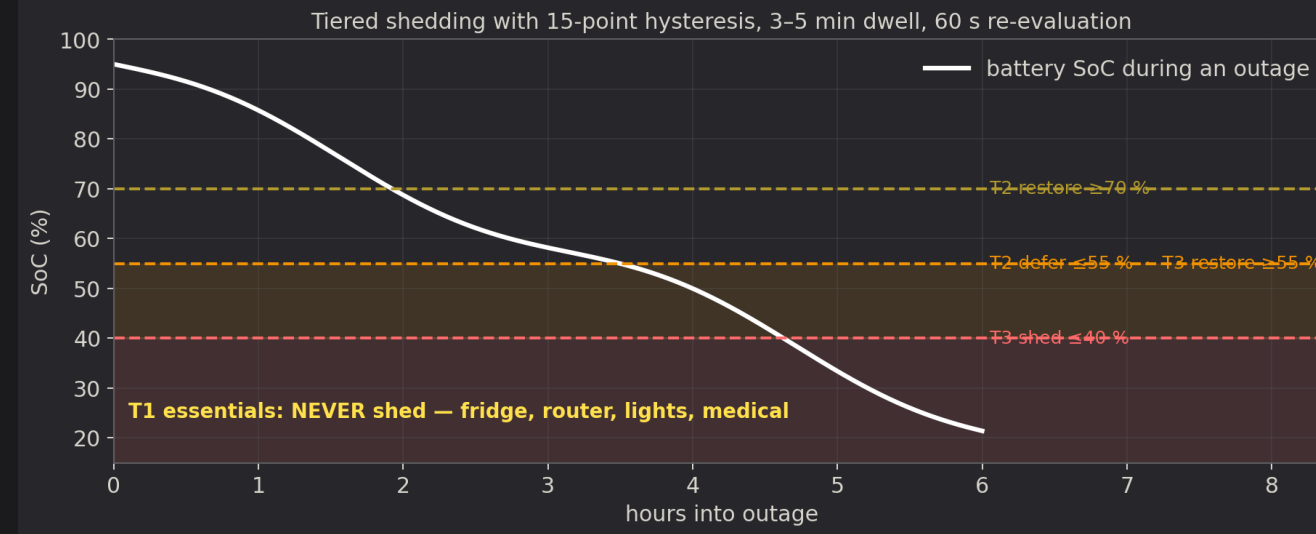
Each outage cycle becomes 14 physics features plus 6 lifetime integrals: resistance and sag vs the battery's own baseline, throughput, depth of discharge, coulombic efficiency, charge acceptance, dQ/dV , Arrhenius heat stress.

03

Infer

An int8 1-D CNN over the last 30 cycles, three seeds averaged, quantile heads P10/P50/P90 with conformal calibration — a window, never a date. 16 KB of RAM, ~1–10 ms per inference on the ESP32-S3.

Measured today (synthetic tubular data, independent test split): SoH MAE 8.2 pt · 80 % band coverage 0.66 → 0.99 after calibration but still 34 pt wide · EKF within ± 3 % vs 7 %/week naive drift · int8 costs +0.01 pt · target ≤ 3 pt needs the aging campaign.



Engine 2: Autopilot That Fails Safe

How It Works

- Hour-of-week load and outage tables (4.7 KB) a judge can read
- Forecasts the roster, not the storm: scheduled cuts are predictable, faults are detected
- Outage declared by a 2-of-3 vote in under 2 s; restore after 10–30 s above 0.9 pu
- T1 never shed · T2 defer $\leq 55\%$ · T3 shed $\leq 40\%$, 15-point hysteresis, 3–5 min dwell

Key Benefits

- Fridge, router, lights and medical loads last longest
- No chatter, no compressor short-cycling: 65 tests, 50/50 scripted scenarios in C
- 30-minute user override that cannot beat the safety floor
- Works with zero internet and no inverter cooperation

Fail-safe by wiring, not by software

Normally-closed contactors: an energised coil sheds, a de-energised coil means loads on. Any crash, brown-out or watchdog reset — and an independent supervisory timer on the driver board — drops every coil within 2 s.

Engine 3: Energy Coach & Grid Shield

How the ESP32 tells appliances apart from one aggregate signal — and why the report's first bill of materials could not do it without a metering front-end:

1 · Detect the step

Moving-average change test on real and reactive power at 3 Hz: $\Delta P \geq 25 \text{ W}$, settle-time limit, ON/OFF pairing.

2 · Sign it

13 numbers per event — ΔP , ΔQ , phase angle, inrush ratio, settling time, harmonic share, duration, periodicity, time of day.

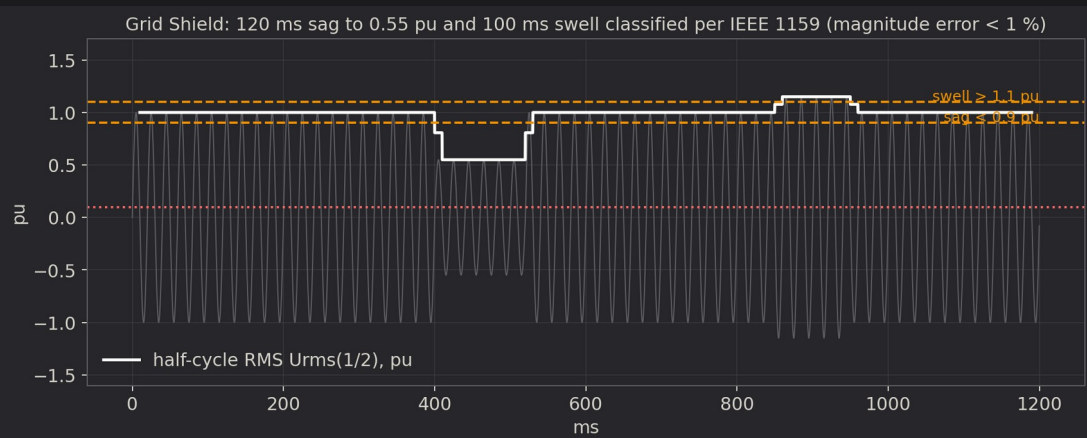
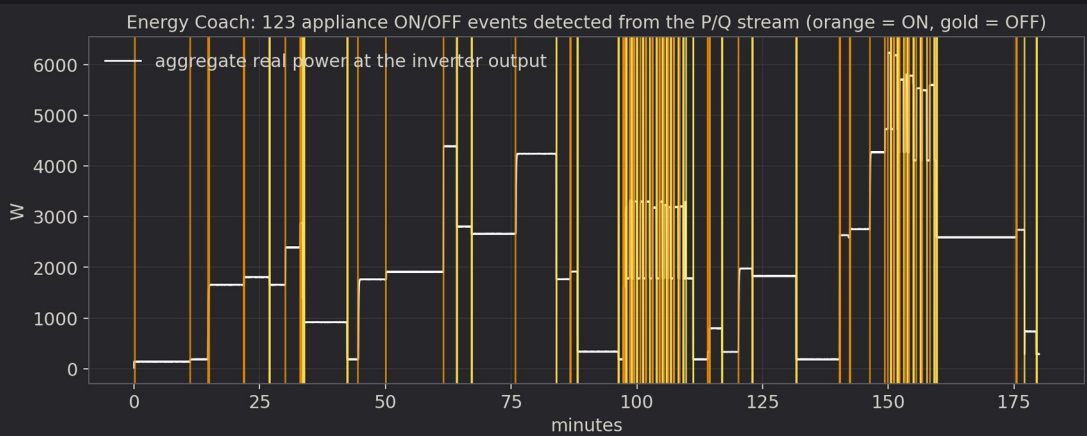
3 · Match it

Rules + k-NN against a per-home library seeded from an Indian appliance table and named by the user: “what just turned on?”

Grid Shield

Half-cycle RMS on an isolated tap, IEEE 1159 buckets, 4-cycle FFT for THD. Class-S-like, never “Class A”.

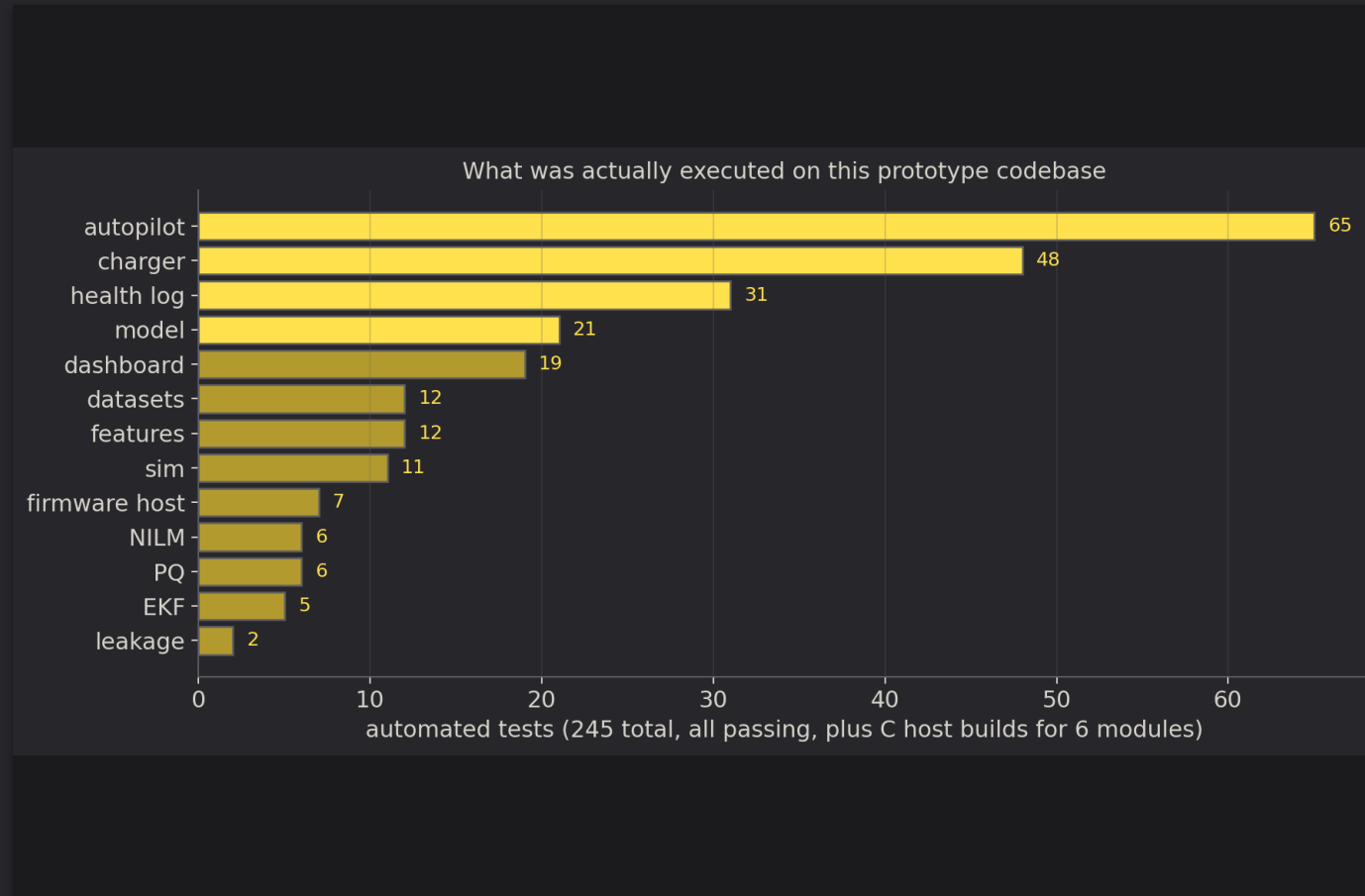
Measured: event recall 1.00, fridge F1 0.97; 13/13 dips detected, magnitude error < 1 %, duration error 0 half-cycles. Honest scope: fridge, AC, geyser, pump, iron/kettle — not two identical fans, not loads under 40 W.



The Prototype: What Actually Runs

✓ 245 automated tests · 6 C modules bit-identical to Python · full-int8 model on 3 seeds · firmware host build

Not a slide deck — a codebase. Every engine implemented, tested and reproducible with one script; the same C is linked into the ESP-IDF firmware.

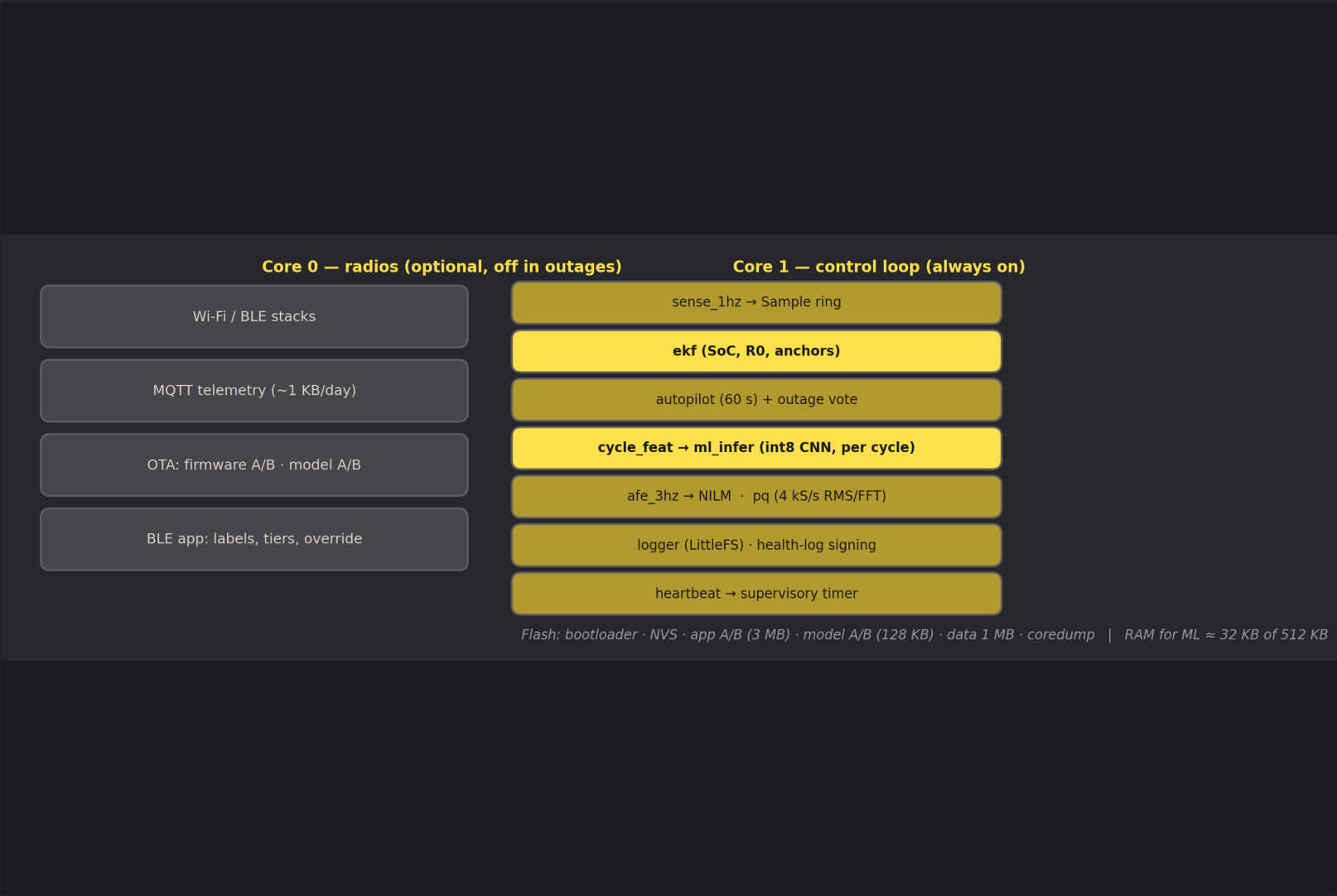


Inside the repository

- **Simulator + features:** synthetic Indian tubular duty cycles at 1 Hz → 14 + 6 physics features per cycle
- **Model pipeline:** train → evaluate → int8 → .tflite, independent test split, leakage guard, transfer ablation
- **EKF · autopilot · NILM · PQ · health log · charger policy:** Python reference + C99 port
- **Firmware:** ESP-IDF task map; host binary runs the loop on a battery CSV and feeds the dashboard
- **Dashboard:** Flask + one page; scripted finale sequence or live device file

Self-Contained by Construction

Unplug the router for a year: every function below still works. The cloud only adds updates and fleet learning.



How the TinyML actually runs

- Fixed int8 weights (3×36.7 KB) in a signed model partition, updated independently of firmware
- TFLite Micro with six registered ops and a 16 KB arena; golden self-test at boot, fallback slot on failure
- One inference per outage cycle: the last 30 cycles in, three quantiles out
- Plain C turns quantiles into weeks and “replace within N weeks” with hysteresis
- No training on the device, ever — personalisation is ratio-to-baseline features and a conformal offset

■ **Draw:** ≈ 2.6 mA average from the battery — 0.04 % of a 150 Ah battery per day.

Live Demonstration



Demo Concept

"A real 12 V tubular battery and inverter on the table, three relay-switched load tiers, nothing connected to the internet — and we pull the processor's reset while the heater circuit is shed to prove that loads come back."

Real vs replayed

- **Real:** SoC and R0 from the shunt and Kalman filter, outage detection, tiered shedding, fail-safe restore, appliance events, signed-log verification
- **Replayed, labelled on screen:** the SoH walk to "Replace within N weeks" — a bench battery does not age in a week

REPOSITORY (PRIVATE): github.com/epshukla/vguard-sentinel — design register (17 decisions), prototype code, run_all.sh reproduces every number on this deck

Proven · Synthetic · Still Needs Hardware

We separate what is deployable today from what is a research target — credibility is the whole game.

PROVEN HERE

245 tests · C ports bit-identical to Python
int8 model exported and self-tested
fail-safe logic, PQ & NILM detectors
signed log with checkpoints

SYNTHETIC ONLY

SoH MAE 8.2 pt on an independent test split
RUL band calibrated (0.99 coverage, too wide)
EKF $\pm 3\%$ vs 7 %/week naive drift
all on simulated tubular batteries

NEEDS HARDWARE / DATA

ESP32 flash + latency on silicon
real battery, CT, PZEM on the bench
aging campaign → real SoH accuracy
life-extension % measured, not assumed

■ **What building it taught us:** float voltage is not open-circuit voltage; a battery shunt cannot see appliances; adaptive charging is embedded-only; federate the model that has labels on the device. Every correction is a numbered decision (D1–D17) in the design register.

Feasibility & Growth Roadmap

Technical Feasibility

- ESP32-S3 with vector extensions, ₹300
- Metering AFE class used in smart meters
- Retrofit needs no inverter modification
- Embedded slots into the Smart Pro family

Validation Targets

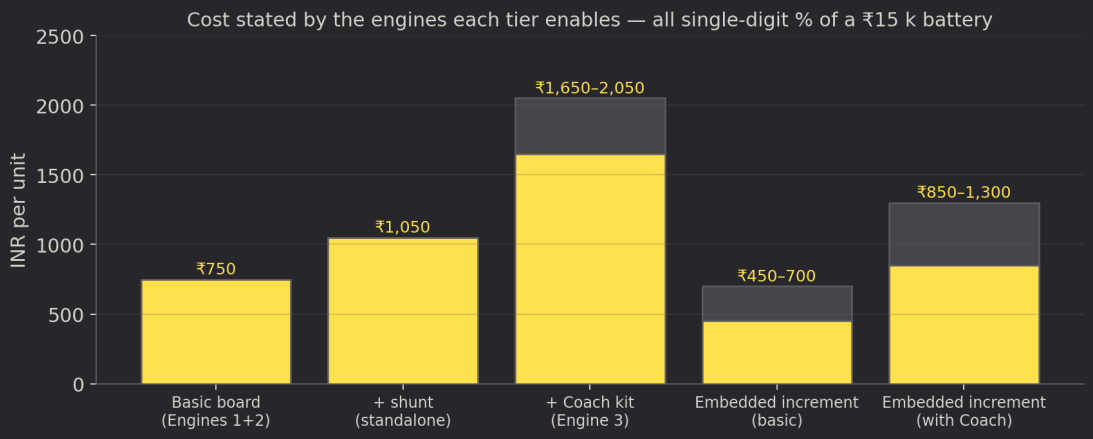
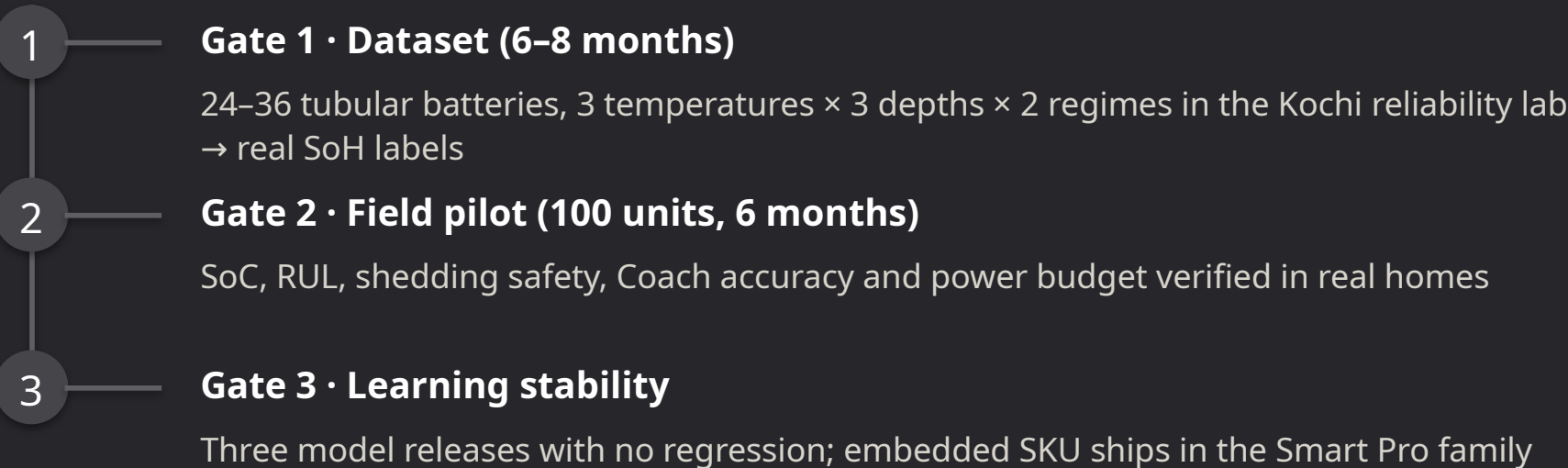
- SoC within 3 % on reference shunts
- ≥ 75 % RUL-window hit-rate
- Zero unsafe sheds; fridge/AC F1 ≥ 0.8
- Life-extension % measured, not assumed

Business Value

- Timely battery replacement revenue
- Warranty decided from signed evidence
- A premium “intelligent” tier
- A fleet-data moat no competitor can buy

What We Need

- ESP32-S3 bench hardware
- One opened Prime-series inverter + charger schematic
- Batteries and cyclers for Gate 1



Coil off means loads on.
A window, never a date.
Offline first, fleet second.

V-Guard Sentinel — Team Codey Tingle (TI3271)

Big Idea Tech 2026 · Track 4 · Finale, Kochi, 24 September 2026

